

Auto-Encoders & Variational Auto-Encoders (VAE)

Funnel approach with technical & mathematical details

Kheireddin Kadri

Lecture Notes

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- Understand the **core idea** of Auto-Encoders (compression $\rightarrow$ reconstruction).
- Clearly distinguish **deterministic AEs** from **probabilistic VAEs**.
- Master the **loss functions** (reconstruction, regularization, ELBO).
- Understand the key mechanism: **the reparameterization trick**.
- Learn how to **diagnose common failures** (collapse, over-regularization, blur).

Hands-on part (later) A Jupyter notebook on **MNIST**: AE $\rightarrow$ Denoising AE $\rightarrow$ VAE, latent visualization and image generation.

Funnel approach (guiding narrative)

- ① **Compression:** why learn compact representations?
- ② **Auto-Encoder:** encoder/decoder + reconstruction.
- ③ **Regularization:** avoid trivial identity solutions.
- ④ **Probabilistic modeling:** we want to *generate*.
- ⑤ **VAE:** generative model + variational inference + ELBO.

Key vision AE = **learning a useful code**. VAE = **learning a generative latent space**.

Core problem: representation and reconstruction

Data $\mathbf{x} \in \mathbb{R}^D$. We seek a representation $\mathbf{z} \in \mathbb{R}^d$ with $d \ll D$.

$$\text{Encoder } f_{\theta} : \mathbb{R}^D \rightarrow \mathbb{R}^d, \quad \mathbf{z} = f_{\theta}(\mathbf{x})$$

$$\text{Decoder } g_{\phi} : \mathbb{R}^d \rightarrow \mathbb{R}^D, \quad \hat{\mathbf{x}} = g_{\phi}(\mathbf{z})$$

Goal: $\hat{\mathbf{x}} \approx \mathbf{x}$.

Intuition A bottleneck forces the network to preserve only the **essential information**.

Loss function (classical AE)

Typical objective:

$$\mathcal{L}_{AE}(\theta, \phi) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\ell(\mathbf{x}, \hat{\mathbf{x}})] \quad \text{with } \hat{\mathbf{x}} = g_{\phi}(f_{\theta}(\mathbf{x})).$$

Common choices:

- **MSE** (real-valued data)
- **BCE** (binary or normalized pixels)

Technical note The loss implicitly defines a **noise model** for $p(\mathbf{x}|\mathbf{z})$.

With linear encoder/decoder and MSE:

$$\min_{W_1, W_2} \mathbb{E} \|\mathbf{x} - W_2 W_1 \mathbf{x}\|^2$$

The solution spans the same subspace as **PCA**.

Takeaway Nonlinear AEs generalize PCA to curved manifolds.

Trivial solutions and regularization

If capacity is too large:

- the AE may learn the identity map

Countermeasures:

- Bottleneck
- Denoising Auto-Encoder
- Sparse Auto-Encoder
- Contractive Auto-Encoder

Key message An AE must be **constrained** to be meaningful.

Denoising Auto-Encoder

Corrupt input:

$$\tilde{\mathbf{x}} \sim q(\tilde{\mathbf{x}}|\mathbf{x})$$

Train to reconstruct $\mathbf{x}$ from $\tilde{\mathbf{x}}$:

$$\mathcal{L}_{DAE} = \mathbb{E}_{\mathbf{x}, \tilde{\mathbf{x}}}[\ell(\mathbf{x}, g_{\phi}(f_{\theta}(\tilde{\mathbf{x}})))]$$

Interpretation The model learns **robust representations** and implicit denoising.

Why AEs are not generative

Latent codes of AEs:

- are not guaranteed to follow a simple distribution
- sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ often produces invalid outputs

Goal of VAE Impose a structured latent distribution to enable **sampling and generation**.

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I), \quad p_{\phi}(\mathbf{x}|\mathbf{z})$$

Maximize:

$$\log p_{\phi}(\mathbf{x}) = \log \int p_{\phi}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

Intractable $\Rightarrow$ variational inference.

Key idea Approximate $p(\mathbf{z}|\mathbf{x})$ with $q_{\theta}(\mathbf{z}|\mathbf{x})$.

$$q_{\theta}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_{\theta}(\mathbf{x}), \text{diag}(\sigma_{\theta}^2(\mathbf{x})))$$

Encoder outputs:

$$\mu(\mathbf{x}), \log \sigma^2(\mathbf{x})$$

Why log-variance? Numerical stability and positivity.

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q_{\theta}(\mathbf{z}|\mathbf{x})}[\log p_{\phi}(\mathbf{x}|\mathbf{z})] - \text{KL}(q_{\theta}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

Interpretation Reconstruction + latent regularization.

KL divergence (closed form)

$$\text{KL} = \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Implementation Exact, no Monte Carlo needed.

Reparameterization trick

$$\mathbf{z} = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Essential Enables backpropagation through sampling.

AE vs VAE (summary)

	AE	VAE
Latent	deterministic	probabilistic
Generative	no	yes
Loss	recon	recon + KL
Sampling	difficult	natural

Conclusion Use AE for representations, VAE for generative modeling.

Next step

MNIST notebook:

- AE, DAE, VAE
- Latent visualization
- Sampling and interpolation

Ready when you are Tell me: **MLP** or **CNN** first?